



As your system scales to higher numbers, it becomes difficult to maintain it

ability is a necessity for it. One way of doing this is to modify the application as and when required to ensure we are able to handle just enough work. This however requires constant development and maintenance effort and might also end up degrading the quality of both the code and the design. Scalability of an application should be handled right from the start, as a non-functional requirement, it is often less prioritised than say, a new shiny app but once the volume of work starts exceeding the application's capabilities, it needs to be scaled immediately.

The correct way of ensuring that an application is scalable is to build it right into the design. Some popular ways of doing this are:

Decreasing the processing time

It's elementary really, if each process of work takes less time to complete, then more amount of it can be done sooner. When designing an application we often strive to achieve perfectly abstracted and modular components. While this is advantageous for a fair number of reasons, sometimes it can be unnecessary and affect performance. Abstractions and data transformations all take up precious processing time, so we must be careful not to overuse them.

Partitioning work

Another way of getting more things done is to partition the work into inde-